

Colorado Medicaid EDI / 837P Developer & Implementation Guide

NEMT direct electronic claim submission roadmap
Prepared for RedArt development and operations

Purpose: Move eligible Colorado Medicaid NEMT billing from browser automation toward a reliable X12 837 Professional (837P) batch EDI workflow, while keeping the existing portal robot as a fallback during onboarding and testing.

Important: This is an engineering roadmap, not a substitute for HCPF/Cotiviti enrollment approval or the licensed ASC X12 implementation specification. Colorado is transitioning EDI infrastructure in 2026, so production connection details must be confirmed with HCPF.

1. Executive Summary

Colorado supports electronic NEMT billing using the HIPAA ASC X12 **837 Professional (837P)** transaction. RedArt's long-term architecture should therefore be a real EDI system rather than a robot that imitates a human filling out the Provider Portal.

Target flow

```
REDART DATABASE
-> Claim Validation Engine
-> X12 837P Generator
-> X12 / Colorado Validation
-> Batch + File Naming
-> EDI / Managed File Transfer
-> Colorado Medicaid
-> Acknowledgements / Remittance
-> RedArt Claim Status
```

Keep the current browser robot operational until RedArt completes Trading Partner onboarding/testing and demonstrates accepted test transactions. Then gradually route eligible claims through EDI.

2. What RedArt Needs From Colorado

- **Trading Partner enrollment / authorization** for RedArt's intended role.
- A Colorado **Trading Partner ID (TPID)** or confirmation of the correct submitter relationship.
- Current 2026 **test credentials** and Managed File Transfer / EDI connection instructions.
- Authorization for transaction sets, especially **837P** and later **835**.
- Confirmation of the current Cotiviti/Edifecs transition requirements.
- Provider authorization/association rules if RedArt submits for multiple transportation providers.

Do not let engineering guess production hostnames, usernames, directories, receiver IDs, or enrollment relationships. Obtain them from current HCPF onboarding.

3. Core Transaction

Primary claim transaction: **HIPAA ASC X12 005010X222A1 - Health Care Claim: Professional - 837P.**

Conceptual hierarchy

```
ISA / IEA - Interchange
GS / GE - Functional group
ST / SE - 837P transaction
BHT - Transaction beginning
NM1/PER - Submitter / receiver
HL - Billing provider / subscriber hierarchy
NM1 - Provider / member identities
CLM - Claim
HI - Diagnosis
LX - Service line
SV1 - Professional service
DTP - Date of service
```

Use a proper X12 data model/generator. Avoid a fragile string-replacement template.

4. Test vs Production

RedArt must have physically and logically separate **TEST** and **PRODUCTION** configurations.

```
TEST
environment = test
ISA15 = T
transport_profile = colorado_test

PRODUCTION
environment = production
ISA15 = P
transport_profile = colorado_production
```

Control the environment centrally. Users should not manually edit ISA15 on individual claims.

5. Do Not Hard-Code Gainwell

Colorado's EDI infrastructure is transitioning in 2026. Build a configurable transport layer so endpoints and credentials can change without rewriting the 837P engine.

```
edi_vendor
edi_host
edi_port
edi_protocol
edi_username
credential_secret_ref
inbound_path
outbound_path
environment
trading_partner_id
receiver_id
active
```

Keep credentials in a secrets manager, not source code or ordinary database text fields.

6. Provider Billing Profiles

RedArt's Trading Partner identity and each transportation provider's billing identity are different. Do not use one global NPI for every customer.

```
provider_billing_profiles
id
company_id
legal_name
billing_name
npi
taxonomy_code
medicaid_provider_id
address_1
address_2
city
state
zip
required_billing_identifiers
trading_partner_relationship
active
```

SUBMITTER != BILLING PROVIDER. RedArt may be the EDI submitter while the transportation company is the billing provider.

7. NEMT Pre-Submission Validation

RedArt should reject incomplete or obviously invalid claims before generating X12.

- Member Medicaid identifier exists and is validly formatted.
- Billing provider identifiers are present.
- Procedure/service code is present and valid for the configured workflow.
- Required diagnosis is present. HCPF NEMT guidance identifies **R68.89** for NEMT billing.
- From and To dates of service are present.
- For a single-day service, From DOS and To DOS match.
- Do not create prohibited transportation span billing.
- Units/mileage are present and internally consistent.
- Charge amount is valid.
- Required place-of-service and service-line data are present.
- Provider/payer routing is known before submission.

```
CANNOT SUBMIT
- Medicaid Member ID missing
- Billing NPI missing
- Mileage cannot be zero
- Required diagnosis missing
- Procedure code missing
- Service date missing
```

8. Batch Model

EDI should be batch-oriented. A batch can contain multiple claims and needs its own lifecycle, file, acknowledgements, and audit information.

```
edi_batches
id
batch_number
provider_id
trading_partner_id
environment
transaction_type
```

```
claim_count
total_charge_amount
file_name
file_hash
generated_at
uploaded_at
status
isa_control_number
gs_control_number
ack_received_at
created_at
updated_at

edi_batch_claims
batch_id
claim_id
transaction_control_number
status
last_response_code
last_response_at
```

9. File Naming

Implement the exact current HCPF filename specification. Current guidance uses TPID, transaction type, timestamp, and file-part information.

```
--<1of1>.
```

Example only:

```
123456789-837P-20260828145130985-1of1.txt
```

Do not put arbitrary RedArt batch IDs into Colorado-reserved filename components. Keep RedArt IDs in the database/audit trail.

10. X12 Control Numbers

Build a concurrency-safe control-number service and store the numbers with the batch/claim.

```
ISA13 - Interchange Control Number
GS06 - Functional Group Control Number
ST02 - Transaction Set Control Number
```

These values are essential for reconciling acknowledgements back to the exact submission.

11. Correct Claim Status Model

A successful upload does **not** mean Medicaid accepted or paid the claim.

```
DRAFT
-> VALIDATING
-> READY
-> 837 GENERATED
-> UPLOADING
-> UPLOADED
-> AWAITING ACKNOWLEDGEMENT
-> EDI ACCEPTED
-> PROCESSING
-> PAID / DENIED / ADJUSTED
```

```
Failure branches:
VALIDATION FAILED
UPLOAD FAILED
SUBMISSION UNKNOWN
```

EDI REJECTED
CLAIM DENIED

The UI should distinguish transport success, EDI acceptance, adjudication, and payment.

12. 999 Functional Acknowledgement

RedArt should automatically download, store, parse, and associate the 999 acknowledgement with the original interchange/transaction.

Batch RB-2026-10048
Claims: 42
837 Uploaded: Aug 28, 2026 2:41 PM
999: ACCEPTED
EDI Status: Accepted for processing

Rejected example:
999: REJECTED
Reason: X12 syntax / segment / element failure
Affected transaction: ST control 000042

An accepted 999 means EDI/structural acceptance; it does not guarantee claim payment.

13. 835 ERA Parser

After 837P submission is stable, add an **835 Electronic Remittance Advice** parser to automate payment and denial reconciliation.

Potential outputs:
Billed amount
Allowed amount
Paid amount
Payment date
EFT/check reference
Claim status
Adjustment groups/reason codes
Service-line payment information

This moves RedArt from claim submission toward a broader billing/revenue-cycle platform.

14. Automatic Response Downloader

Build a scheduled inbound EDI worker. Retrieve response files promptly and follow the current HCPF retention rules.

```
connect to configured endpoint
-> list inbound files
-> identify unseen files
-> download securely
-> hash file
-> store audit metadata
-> parse response type
-> reconcile batch/claim
-> update status
-> record processing result
```

15. Idempotency & Duplicate Protection

Inbound and outbound operations must be safe against duplicate execution.

remote_filename
remote_file_id (if available)
file_hash

```
downloaded_at
processed_at
processing_status
parser_version
```

If the same file/hash is seen again, do not post the same acknowledgement/remittance twice.

16. Never Blindly Retry Unknown Submission

If a network timeout happens after upload starts, RedArt may not know whether Colorado received the batch.

```
status = SUBMISSION_UNKNOWN
```

Before resending, reconcile remote listings, acknowledgements, control numbers, hashes, and audit history. Blind retries can create duplicate submissions.

17. Audit Trail & PHI-Safe Logging

- Original generated 837 or securely retrievable immutable copy.
- File hash and generation timestamp.
- Upload attempts and final transport result.
- RedArt batch ID and included claim IDs.
- Trading Partner / billing provider relationship.
- ISA, GS and ST control numbers.
- 999 and other acknowledgement files.
- 835/remittance data when implemented.
- User/system actor that triggered submission.
- Application and parser/generator version.

Avoid logging complete X12 payloads or unnecessary member PHI to ordinary application logs. Log batch IDs, hashes, control numbers, status codes, and sanitized errors.

18. EDI Diagnostics UI

```
EDI DIAGNOSTICS
Environment: TEST
Trading Partner: masked TPID
Connection: Connected
837P Generator: Healthy
Last Upload: 2:42 PM
Last Response Poll: 2:45 PM
Last 999: Accepted
Pending Batches: 2
Rejected Batches: 0
```

Tabs:

Batches | Acknowledgements | 835 Remittances | Errors | Connections | Audit

19. Provider / Payer Routing

Do not assume every NEMT trip belongs in the same destination. Make reimbursement/billing routing configurable.

```
Trip / bill
-> determine payer/network/billing route
-> Colorado Medicaid EDI
-> configured broker workflow
-> other configured payer/workflow
```

Store the routing decision on the claim before submission so a claim cannot accidentally go to the wrong channel.

20. Recommended Development Milestones

- **M1:** Colorado EDI configuration and provider billing profile schema.
- **M2:** NEMT pre-submission validation.
- **M3:** Generate one structurally correct 837P locally.
- **M4:** Control numbers, batch model, hashes and audit records.
- **M5:** Current Colorado filename rules.
- **M6:** Configurable TEST transport.
- **M7:** Generate/send the HCPF-required test claim set.
- **M8:** Automatically retrieve and parse 999.
- **M9:** Achieve accepted test transactions and production authorization.
- **M10:** Controlled production pilot with a very small batch.
- **M11:** Expand eligible claims gradually.
- **M12:** Add 835 ERA reconciliation.

21. Developer Handoff Specification

REDART COLORADO EDI PROJECT

Goal:

Replace browser-based Colorado Medicaid claim submission with direct X12 batch submission where permitted.

Transaction:

HIPAA ASC X12 005010X222A1 - 837 Professional

Primary use:

Colorado NEMT claims

Build:

1. Trading Partner configuration
2. Provider billing profiles
3. NEMT claim validator
4. X12 837P generator
5. ISA/GS/ST control number manager
6. Test/Production separation
7. Colorado file naming
8. Configurable MFT/SFTP transport
9. Submission batch management
10. 999 acknowledgement parser
11. 835 ERA parser
12. Automatic response downloader
13. Claim/batch reconciliation
14. Duplicate prevention / idempotency
15. Complete EDI audit trail
16. EDI diagnostics UI
17. Encryption / PHI-safe logging
18. Provider / payer routing layer

IMPORTANT:

- Do not hard-code the old Gainwell environment.
- Do not mark a claim Submitted merely because upload succeeded.
- Do not blindly retry an uncertain upload.
- Do not send production transactions before authorization.
- Use Colorado's Companion Guide plus the official ASC X12 005010X222A1 specification.

22. Official Resources

HCPF EDI Support:

<https://hcpf.colorado.gov/edi-support>

Colorado NEMT Billing Manual:

<https://hcpf.colorado.gov/nemt-billing-manual>

Colorado 837P Companion Guide:

https://hcpf.colorado.gov/sites/hcpf/files/CO_EDI_v5010_X12_837P_CompanionGuide%20-%202012-18-2023.pdf

Trading Partner Testing Packet:

<https://hcpf.colorado.gov/sites/hcpf/files/Trading%20Partner%20Testing%20Packet%20-%202003272023.pdf>

Colorado EDI File Naming Requirements: use the latest version published from the HCPF EDI Support page.

ASC X12 005010X222A1 implementation specification: obtain/use the official licensed X12 implementation guide as required.

23. Recommended Next Action

Business/onboarding side: confirm RedArt's Trading Partner status/TPID, current 2026 onboarding path, test credentials, provider authorization model, and current Cotiviti/Edifecs MFT requirements.

Engineering can start immediately on the database model, validator, X12 abstraction, batch/control-number system, audit trail, and TEST-only generator while onboarding is being completed.

Do not remove the existing billing robot yet. Keep it as a fallback until RedArt has successfully completed testing and production EDI is proven.